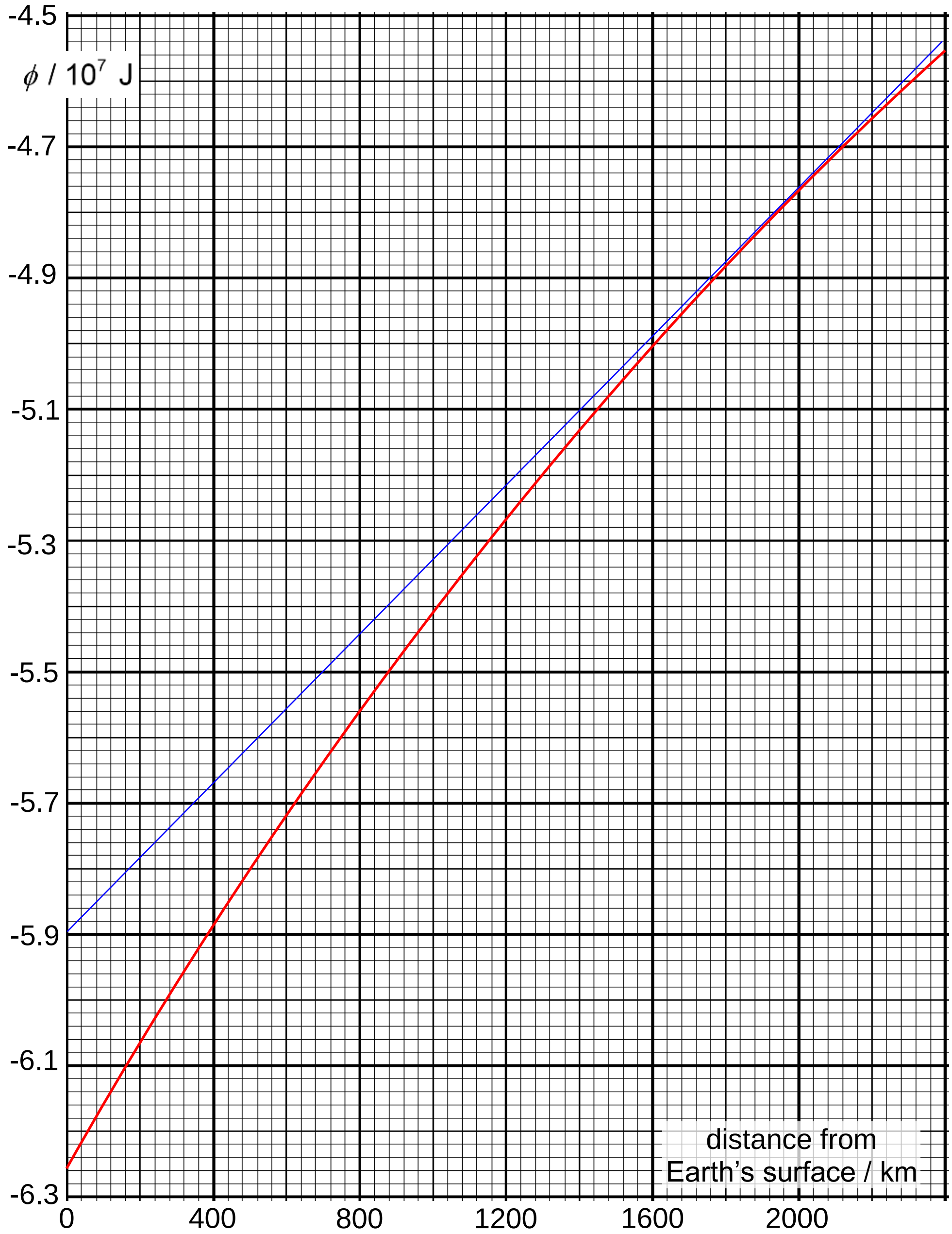
Qns	Answer	Marks
8(a)(i)	$v_y = v\cos\theta$ $= (536)\cos(90° - 86.7°)$ $= 535 \text{ m s}^{-1}$	 1 A0
(a)(ii)	$s = ut + \frac{1}{2}at^2$ $-50.4 = (535)t + \frac{1}{2}(-9.81)t^2$ $t = 109 \text{ s}$	 1 1
(a)(iii)	$v^2 = u^2 + 2as$ $0 = 535^2 + 2(-9.81)s_{max}$ $s_{max} = 14588 \text{ m}$ $\text{height} = 14588 + 50.4$ $\approx 14600 \text{ m}$	 1 1
(b)(i)	$\frac{\cancel{mg}h_{actual}}{\cancel{mg}h_{ideal}} = \frac{9300}{14600} = 0.637$	1
(b)(ii)	Larger Drag force increases with larger <u>relative velocity</u>. Traditional rockets start off stationary and hence experience less drag across the ascend process	1 1
(c)(i)	$\omega = \frac{2\pi}{T} = \frac{2\pi}{24 \times 60 \times 60} = 7.27 \times 10^{-5} \text{ rad s}^{-1}$	1
(c)(ii)	Towards east. Launching projectile in the same direction as Earth's rotation reduces the amount of kinetic energy that the accelerator has to transfer to the projectile for the projective to achieve the desired speed.	1 1
(d)(i)	$v = r\omega = r\left(\frac{\theta}{t}\right)$ $= (50.0)\left(\frac{2\pi(450)}{60}\right)$ $= 2356$ $= 2360 \text{ m s}^{-1}$	 1

Qns	Answer	Marks
(d)(ii)	$v_{\text{at launch}} = \dfrac{8000\times10^3}{60\times60} = 2220 \text{ m s}^{-1}$ $\text{ratio} = \dfrac{\cancel{\frac{1}{2}}\cancel{m}v^2_{\text{at launch}}}{\cancel{\frac{1}{2}}\cancel{m}v^2_{\text{before}}} = \left(\dfrac{2220}{2360}\right)^2 = 0.880 \ \ (\text{accept } 0.890)$	**1**
(d)(iii)	**Minimise loss of** (kinetic) **energy** as heat due to work done against **air resistance**	**1**
(e)(i)	$\text{KE just before launch} = \dfrac{1}{2}mv^2$ $= \dfrac{1}{2}(200)(2360)^2$ $= 5.57\times10^8 \text{ J}$ $\text{time needed} = \dfrac{5.57\times10^8}{100\times10^3} = 5570 \text{ s} = 92.8 \text{ min}$	
(e)(ii)	$\dfrac{a_c}{g} = \dfrac{r\omega^2}{g}$ $= (50.0)\left(\dfrac{450(2\pi)}{60}\right)^2 \dfrac{1}{9.81}$ $= 11300$ No. They were subject to more than 10 000 *g* for more than an hour.	**1** **1**
(f)(i)	**Work by per unit mass** in bringing **a small test mass** from **infinity to that point**	**1**

<table>
<tr><th>Qns</th><th>Answer</th><th>Marks</th></tr>
<tr><td>(f)(ii)</td><td>



$$g = -\frac{\mathrm{d}\phi}{\mathrm{d}r} \approx \frac{\left[(-4.54)-(-5.9)\right]\times 10^7}{(2400-0)\times 10^3} = 5.67\ \text{N kg}^{-1}$$

</td><td>1
1</td></tr>
<tr><td>(f)(iii)</td><td>The astronauts experience the same acceleration towards centre of the Earth as the space craft.

Hence there is no contact force by space craft on the astronaut.</td><td>1

1</td></tr>
<tr><td>(f)(iv)</td><td>

$$\Delta U = m_{\text{projectile}}(\Delta\phi) = m_{\text{projectile}}(\phi_{\text{final}} - \phi_{\text{initial}})$$
$$= (200)\left(\left(-5.89-(-6.25)\right)\times 10^7\right)$$
$$= 7.20\times 10^8\ \text{J}$$

</td><td>1
1</td></tr>
</table>